

— GENOMICS & DATA SCIENCE · 3-MONTH INTENSIVE

Bioinformatics *Bootcamp*

Learn to analyse the data of modern biology, from raw sequences to genomic insight, across computational biology, genomics and data-driven health research.

DURATION

3 Months

FORMAT

Onsite & Online

FEE

NGN 300,000

— THE PROGRAMME

Where biology meets data science

This three-month programme is tailored for students, researchers and professionals in the biological sciences who want to harness data in modern research. You will master the tools and techniques used to analyse biological data, with a strong focus on computational biology, genomics and data-driven health research.

Data-Lead Africa is a research and training consulting firm working across monitoring & evaluation, data analytics, data science training and strategic consulting. We combine statistics, research methods and technology to turn data into insight, taught by a team of seasoned industry experts.

Who this is for

Biology students

Researchers

Life scientists

Healthcare professionals

Biotech professionals

Aspiring bioinformaticians

PARTNERS & CLIENTS WE HAVE WORKED WITH

USAID
U.S. Agency for International DevelopmentMERCY
CORPSjhpiego
Johns Hopkins University AffiliateSave the
Childrenjica
Japan International
Cooperation AgencyACTION
AGAINST
HUNGERACTION
CONTRE
LA FAIMUNITED NATIONS TRUST FUND
TO END VIOLENCE AGAINST WOMENNJR
Nigeria Joint Response
Unit Relief Alliance

— WHY IT MATTERS

From *raw sequence* to *real insight*.

Biology now produces data faster than anyone can read it. Bioinformatics gives you the computational skills to turn sequences, genomes and experiments into discoveries that move research and health forward.

THE SKILLS YOU WILL GAIN

Genomic data

Command line

Sequence alignment

NGS analysis

Biostatistics

Machine learning

Data visualisation

Reproducible pipelines

WHAT YOU WILL WALK AWAY ABLE TO DO

- Handle genomic data and work confidently on the command line
- Align sequences, annotate genes and search biological databases
- Apply statistics and machine learning to biological data
- Program bioinformatics analyses in Python and R
- Analyse NGS data, from quality control to variant calling and RNA-seq

— HOW IT WORKS

Structured, hands-on training

Four ways we take you from learning to doing.

Recorded video lessons

High-quality, easy-to-follow lessons you can revisit any time.

Onsite and online

Learn and collaborate with peers, in person or remotely.

Peer-review sessions

Trade perspectives and sharpen your problem-solving.

Real-world case studies

Apply every skill to authentic HR and business problems.

DURATION	FORMAT	INVESTMENT	LEVEL
3 Months	Onsite & Online	NGN 300,000	Basic to advanced

— THE CURRICULUM

What you will master

01**Foundations of Bioinformatics** **BIOLOGY + COMMAND LINE**

Start with the biology and the computing environment every bioinformatician relies on.

The central dogma: DNA, RNA and proteins / Biological data and formats: FASTA, FASTQ, GenBank / Working confidently on the Linux command line

02**Programming for Bioinformatics** **PYTHON & R**

Write the code that automates biological data analysis, from the basics up.

Python and Biopython for sequences and files / R and the Bioconductor ecosystem / Writing clean, reproducible analysis scripts

03**Biological Databases and Data Retrieval** **FINDING DATA**

Find, fetch and parse data from the world's major biological repositories.

NCBI, Ensembl, UniProt and the PDB / Programmatic querying and parsing of records / Genome assemblies and annotation

04**Sequence Alignment and Gene Annotation** **SEQUENCE ANALYSIS**

Compare sequences and work out what genes do.

Pairwise and multiple sequence alignment / BLAST searches and homology / Gene prediction and functional annotation

— THE CURRICULUM, CONTINUED

Model, visualise, apply

05

Genomics and NGS Data Analysis NEXT-GENERATION SEQUENCING

Handle real high-throughput sequencing data end to end.

FASTQ quality control and trimming / Read mapping and variant calling / RNA-seq and differential gene expression

06

Statistical Analysis of Biological Data BIOSTATISTICS

Draw sound, defensible conclusions from complex biological data.

Descriptive and inferential statistics in R / Multiple testing and high-dimensional data / Interpreting significance in omics results

07

Machine Learning for Biology PREDICTIVE BIOINFORMATICS

Use modern machine learning to find patterns in biological data.

Supervised and unsupervised learning / Classification, clustering and feature selection / Biomarker discovery from genomic and clinical data

08

Visualisation and Applications INSIGHT & IMPACT

Communicate results clearly and see where bioinformatics makes a difference.

Volcano plots, heatmaps and genome browsers / Applications: precision medicine, drug discovery, phylogenetics / Reproducible pipelines and sharing results



Applied Capstone

Run an end-to-end bioinformatics project on a real biological dataset, taking it from raw sequence data through analysis to interpreted, visualised results that you present to complete the programme.

— TOOLS, REQUIREMENTS & ENROLMENT

Everything you need to start

TOOLS YOU WILL USE

Python

Biopython

R

Bioconductor

BLAST

Linux

NCBI / Ensembl

Galaxy

Laptop & requirements

Operating system	Windows, macOS or Linux
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Memory (RAM)	8 GB min, 16 GB recommended
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Storage	256 GB SSD
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Processor	Quad-core recommended
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Heavier genomic analyses can run on free platforms such as Galaxy or Google Colab, so you do not need a high-powered computer to begin.

Who should apply

Students, researchers and professionals across biology, biomedical, health and life sciences who want to analyse biological data. Some programming exposure helps but is not required; we start from the basics.

How to enrol

1

Register

Complete the short enrolment form to express your interest.

2

Secure your seat

Make the one-time payment to confirm your place.

3

Onboard and begin

Get your welcome pack, join the cohort and start learning.

ONE-TIME FEE

NGN 300,000

Covers the full three-month programme: video lessons, live sessions, peer reviews, real research case studies and your capstone.



Ready to decode the data of *life*?

Registration is open. Secure your seat in the next Bioinformatics cohort and start turning biological data into discovery.

ENROL NOW

dataleadafrica.com/courses

ONE-TIME FEE

NGN 300,000

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